

GENERAL PROPERTIES OF MANY PARTICLE SYSTEMS AT LOW TEMPERATURES

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CONTENTS

1 The Dilute Fermi Gas

1

1. THE DILUTE FERMION GAS

In the case of the Fermi gas, we determine the ground state energy, the effective mass of the excitations, & the function f , w/ accuracy up to terms of order $(a/\lambda)^2$. Note that a is the amplitude of the s-wave scattering.

We cannot assume the same operator corresponding to the interaction energy using the form in §1.4. The sum will vanish by the anticommutativity of Fermi operators. Additionally, the Hamiltonian in §1.4 doesn't take into account Fermi scattering.

s-scattering can only occur for identical particles of spin-1/2 when the spins are anti-parallel. This makes the amplitude twice of what it is when the particles are different. By these facts, the interaction energy is:

$$H_{\text{int}} = \frac{U}{V} \sum_{p_1+p_2=p_3+p_4} a_{p_3,1/2}^\dagger a_{p_4,-1/2}^\dagger a_{p_2,-1/2} a_{p_1,1/2} \quad (1)$$

letting the subscript of $1/2 = \alpha$ & $-1/2 = \beta$, (1) becomes:

$$H_{\text{int}} = \frac{U}{V} \sum_{p_1+p_2=p_3+p_4} a_{p_3,\alpha}^\dagger a_{p_4,\beta}^\dagger a_{p_2,\beta} a_{p_1,\alpha} \quad (2)$$

Recall that the quantity U is connected w/ the amplitude of s-wave scattering by the same formula from §1.4:

$$U = \frac{4\pi a}{m} \quad (3)$$

We now apply first order perturbation theory to (2). From (2), since $p_1 = p_3$ & $p_2 = p_4$, we have:

$$\begin{aligned} p_1 + p_2 &= p_3 + p_4 \\ p_1 + p_2 &= p_1 + p_2 \quad ; \text{ This is the direct term.} \end{aligned}$$

If we switch the pairings,

$$\begin{aligned} p_4 &= p_1 \quad (\text{requires } \beta = \alpha) \\ p_3 &= p_2 \quad (\text{requires } \alpha = \beta) \quad ; \text{ thus,} \end{aligned}$$

$$p_1 + p_2 = p_3 + p_4$$

$$p_1 + p_2 = p_2 + p_1 \quad ; \text{ This is the fock term.}$$

Each term gets evaluated by the expectation value $\langle \Phi_0 | H_{\text{int}} | \Phi_0 \rangle$. Looking @ the direct term, we have

$$p_3 = p_1 \quad ; \quad p_4 = p_2$$

The operator sequence inside the expectation value is:

$$\langle a_{p_1, \alpha}^\dagger a_{p_2, \beta}^\dagger a_{p_2, \beta} a_{p_1, \alpha} \rangle$$

Our aim is to turn these into number operators. Notice that the middle two operators already meet the aim, as they are adjacent to one another.

$$\begin{aligned} &= \langle a_{p_1, \alpha}^\dagger [a_{p_2, \beta}^\dagger a_{p_2, \beta}] a_{p_1, \alpha} \rangle \\ &= \langle a_{p_1, \alpha}^\dagger n_{p_2, \beta} a_{p_1, \alpha} \rangle \\ &= \langle a_{p_1, \alpha}^\dagger a_{p_1, \alpha} \rangle n_{p_2, \beta} \\ &= n_{p_1, \alpha} n_{p_2, \beta} \quad // \end{aligned}$$

Thus, its contribution is:

$$E^{(1)} = \frac{U}{2V} \sum_{p_1, p_2} n_{p_1, \alpha} n_{p_2, \beta}$$

Repeating this process for the fock term, recall that we set $p_4 = p_1$ & $p_3 = p_2$. The operator sequence is:

$$\langle a_{p_2, \alpha}^\dagger a_{p_1, \alpha}^\dagger a_{p_2, \alpha} a_{p_1, \alpha} \rangle$$

Looking at the innermost two terms first,

$$a_{p_1, \alpha}^\dagger a_{p_2, \alpha} = -a_{p_2, \alpha} a_{p_1, \alpha}^\dagger$$

Supplying this to the operator sequence yields:

$$\begin{aligned} &= \langle a_{p_2, \alpha}^\dagger (-a_{p_2, \alpha} a_{p_1, \alpha}^\dagger) a_{p_1, \alpha} \rangle \\ &= -\langle a_{p_2, \alpha}^\dagger a_{p_2, \alpha} a_{p_1, \alpha}^\dagger a_{p_1, \alpha} \rangle \\ &= -n_{p_2, \alpha} n_{p_1, \alpha} \quad // \end{aligned}$$

Thus, its contribution is:

$$E^{(1)} = -\frac{U}{2V} \sum_{p_1, p_2} n_{p_1, \alpha} n_{p_2, \alpha} \delta_{\alpha\beta}$$

Combining each $E^{(1)}$ term yields:

$$\begin{aligned} E^{(1)} &= \frac{U}{2V} \sum_{i, k} n_i n_k (1 - \delta_{\alpha_i \alpha_k}) \\ &= \frac{U}{V} \sum_{i, k} n_i n_k Q_{ik} \end{aligned} \tag{4}$$

Q_{ik} accounts for anti-parallel spins. This factor can conveniently be written by:

$$Q_{ik} = \frac{1}{4}(1 - \sigma_i \cdot \sigma_k) \quad (5)$$

Substitution of (3) & (5) into (4) yields

$$\begin{aligned} E^{(1)} &= \frac{\frac{4\pi a}{m}}{V} \cdot N^2 \left[\frac{1}{4}(1 - \sigma_i \cdot \sigma_k) \right] \\ E^{(1)} &= \frac{4\pi a}{m} \cdot \frac{N^2}{V} \cdot \frac{1}{4}(1 - \sigma_i \cdot \sigma_k) \\ E^{(1)} &= \frac{\pi a}{m} \frac{N^2}{V} \end{aligned} \quad (6)$$

To find the second order correction, we use the following from perturbation theory:

$$E_s^{(2)} = \sum_{r \neq s} \frac{|H_{\text{int}}|^2}{E_s - E_r} \quad (7)$$

Substitution of (1) into (7) yields:

$$|(H_{\text{int}})_{sr}|^2 = |\langle r | H_{\text{int}} | s \rangle|^2$$

By (1), $a_{p_2, -1/2} a_{p_1, 1/2} = \sqrt{n_{p_1} n_{p_2}}$. Also, $a_{p_3, 1/2}^\dagger a_{p_4, -1/2}^\dagger = \sqrt{(1 - n_{p_3})(1 - n_{p_4})}$. Letting $p_1 \rightarrow i$, $p_2 \rightarrow k$, $p_3 \rightarrow l$, $p_4 \rightarrow m$,

$$\begin{aligned} |\langle r | H_{\text{int}} | s \rangle|^2 &= n_{p_1} n_{p_2} (1 - n_{p_3})(1 - n_{p_4}) \\ &= n_i n_k (1 - n_l)(1 - n_m) \end{aligned}$$

For the denominator, the unperturbed energy of any state is the sum of its single particle kinetic energies: $\sum p^2/2m$.

$$E_r - E_s = \frac{p_i^2 + p_k^2 - p_l^2 - p_m^2}{2m}$$

Putting the numerator & denominator together, along w/ the factor of $\frac{U}{V}$ from (1), we obtain:

$$E_s^{(2)} = \sum_{iklm} \frac{n_i n_k (1 - n_l)(1 - n_m) Q_{ik} Q_{lm}}{(p_i^2 + p_k^2 - p_l^2 - p_m^2)/(2m)} \quad (8)$$

Since we want to find an expression of the energy in terms of scattering amplitude a , we use (3), but note it's only valid up until the 1st order. Taking into account the second order terms,

$$U = \frac{4\pi a}{m} = U + \frac{2U^2}{V} \sum_{k,l} \frac{Q_{kl}}{(p_i^2 + p_k^2 - p_l^2 - p_m^2)/2m} \quad (9)$$

Using (9) to express U in terms of a , & substitution of this result into (4), we find that terms proportional to a^2 appear in $E^{(1)}$. Hence, we should include them into our second-order correction.

$$\begin{aligned}
U &= \frac{4\pi a}{m} - \frac{2U^2}{V} \sum_{k,l} \frac{Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} \\
&= \frac{4\pi a}{m} - \frac{2}{V} \left(\frac{4\pi a}{m} \right)^2 \sum_{k,l} \frac{Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} \\
&= \frac{4\pi a}{m} - \frac{2}{V} \left(\frac{16\pi^2 a^2}{m^2} \right) \sum_{k,l} \frac{Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} \\
&= \frac{4\pi a}{m} - \frac{32\pi^2 a^2}{Vm^2} \sum_{k,l} \frac{Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m}
\end{aligned}$$

Substitution into (4) yields:

$$E^{(1)} = \frac{1}{2V} \left[\frac{4\pi a}{m} - \frac{32\pi^2 a^2}{m^2 V} \sum_{k,l} \frac{Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} \right] \sum_{i,k} n_i n_k Q_{ik}$$

By distribution, we obtain:

$$E^{(1)} = \frac{2\pi a}{mV} \sum_{i,k} n_i n_k Q_{ik} - \frac{16\pi^2 a^2}{m^2 V^2} \sum_{i,k,l,m} \frac{n_i n_k Q_{ik} Q_{lm}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} \quad (10)$$

[renormalized $E^{(1)}$]

Now, $E^{(2)}$ becomes:

$$E^{(2)} \approx \frac{16\pi^2 a^2}{m^2 V^2} \sum_{i,j,k,l} \left[\frac{n_i n_j (1 - n_k)(1 - n_l) Q_{ij} Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} - \frac{n_i n_j Q_{ij} Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} \right] \quad (11)$$

From (11) the term w/ four n_i (in the first summand) = 0 b/c the numerator is antisymmetric under the permutation $i, j \rightleftharpoons k, l$. The remaining two terms of three n_i are equal. This gives

$$E^{(2)} = \frac{-32a^2\pi^2}{m^2 V^2} \sum_{i,j,k,l} \frac{n_i n_j n_k Q_{ij} Q_{kl}}{(p_i^2 + p_j^2 - p_k^2 - p_l^2)/2m} \quad (12)$$

From summation to integration, (12) can be re-written as:

$$E^{(2)} = \frac{-32a^2\pi^2}{m^2 (2\pi)^9} \times \int_{|p_1| < p_F} dp_1 \int_{|p_2| < p_F} dp_2 \int_{|p_3| > p_F} dp_3 \int dp_4 \frac{\delta(p_1 + p_2 - p_3 - p_4)}{(p_1^2 + p_2^2 - p_3^2 - p_4^2)/2m} \quad (13)$$

We use excitation energy from §2:

$$\varepsilon_i = \frac{\delta E}{\delta n_i} \quad (14)$$

Variations of (4) & (12) w/ respect to n_i yield ; From (4) there are two places n_i can appear. If we differentiate w/ respect to n_i , it leaves n_k . If we differentiate w/ respect to n_k , it leaves n_i . This doubles the result, cancelling out the factor of 1/2 that appears in the pre-factor, leaving us w/ the term : $\frac{2\pi a N}{mV}$.

From (12), we are dealing w/ a product of 4 occupation variables in the numerator:

$$n_i n_k (1 - n_l)(1 - n_m) - n_i n_k$$

Expansion yields:

$$\begin{aligned} &= n_i n_k - n_i n_k n_m - n_i n_k n_l + n_i n_k n_l n_m - n_i n_k \\ &= -n_i n_k n_m - n_i n_k n_l + n_i n_k n_l n_m \end{aligned}$$

The product rule is applied to each product. This gives :

$$\begin{aligned} \mathcal{E}(p) &= \frac{p^2}{2m} + \frac{2\pi a N}{mV} + \frac{16\pi^2 a^2}{m^2 (2\pi)^9} \int_{|p_1| < p_F} dp_1 \int_{|p_2| < p_F} dp_2 \int dp_3 \times \\ &\quad \left[\frac{\delta(p_1 + p_2 - p - p_3)}{(p^2 + p_3^2 - p_1^2 - p_2^2)/2m} - 2 \frac{\delta(p_1 - p - p_2 - p_3)}{(p^2 + p_1^2 - p_2^2 - p_3^2)/2m} \right] \end{aligned} \quad (15)$$

Evaluation of (13) & (15) evaluates the ground state energy & the effective mass of the excitations. However, by the looks of it, it could get extremely laborious. Luckily, there is a simpler method:

Let us introduce a function f:

$$f_{ik} = \frac{\delta^2 E}{\delta n_i \delta n_k} \quad (16)$$

Using (16), we can find the effective mass & velocity of low-frequency sound w/ formulas from §2. Integration will then yield the energy of the ground state.

Varying (4) & (12) w/ respect to f, we obtain:

$$\begin{aligned} f &= \frac{8\pi a}{m} Q_{\sigma\sigma'} - \frac{64\pi^2 a^2}{m^2 (2\pi)^3} \int_{|p_1| < p_0} dp_1 \int_{|p_2| < p_0} dp_2 \left[Q_{\sigma\sigma'} \frac{\delta(p + p' - p_1 - p_2)}{(p^2 + p'^2 - p_1^2 - p_2^2)/2m} \right. \\ &\quad \left. + \frac{1}{4} \frac{\delta(p + p_1 - p' - p_2)}{(p^2 + p_1^2 - p'^2 - p_2^2)/2m} + \frac{1}{4} \frac{\delta(p' + p_1 - p - p_2)}{(p'^2 + p_1^2 - p^2 - p_2^2)/2m} \right] \end{aligned} \quad (17)$$

letting $|p| = |p'| = p_0$, we get something simpler to solve. Recall:

$$Q_{\sigma\sigma'} = \frac{1}{4}(1 - \sigma \cdot \sigma') //$$

Let χ be the angle between p & p' . @ the fermi surface where $|p| = |p'| = p_0$, the total incoming momentum magnitude $P = |p + p'|$ can be simplified w/ the half-angle identity:

$$P^2 = p_0^2 + p_0^2 + 2p_0^2 \cos \chi = 4p_0^2 \cos^2 \left(\frac{\chi}{2} \right) //$$

looking @ the first integral over $|p_1| < p_0$:

$$\int_{|p_1| < p_0} \frac{dp_1}{(p_0^2 + p_0^2 - p_1^2 - |p + p' - p_1|^2)/2m} = 2\pi p_0 m \left[1 - \frac{\sin(\chi/2)}{2} \ln \frac{1 + \sin(\chi/2)}{1 - \sin(\chi/2)} \right]$$

Notice that the 2nd & 3rd terms are symmetric under $p \longleftrightarrow p'$; They are identical

$$\frac{2 + \cos \chi}{2 \sin(\frac{\chi}{2})} \ln \frac{1 + \sin(\frac{\chi}{2})}{1 - \sin(\frac{\chi}{2})}$$

Lastly, using $p_0 = \frac{N}{V}$,

$$p_0 = \pi \left(\frac{3N}{\pi V} \right)^{\frac{1}{3}}$$

Combining all these pieces, we obtain:

$$\begin{aligned} f(\chi) = & \frac{2\pi a}{m} \left[1 + 2a \left(\frac{3N}{\pi V} \right)^{\frac{1}{3}} \left(2 + \frac{\cos \chi}{2 \sin(\frac{\chi}{2})} \ln \frac{1 + \sin \frac{1}{2}\chi}{1 - \sin \frac{1}{2}\chi} \right) \right] \\ & - \frac{2\pi a}{m} (\sigma \cdot \sigma') \left[1 + 2a \left(\frac{3N}{\pi V} \right)^{\frac{1}{3}} \left(1 - \frac{\sin \frac{1}{2}\chi}{2} \ln \frac{1 + \sin \frac{1}{2}\chi}{1 - \sin \frac{1}{2}\chi} \right) \right] \end{aligned} \quad (18)$$

There is a special feature in (18) worth addressing. When χ is near π , f has a logarithmic singularity for particles w/ opposite spins.

$$f(\chi) \propto (1 - \sigma \cdot \sigma') \ln \frac{1}{\pi - \chi} \quad (19)$$

If we set $\chi = \pi$ & write:

$$\lambda_P = p^2 + p'^2 - 2p_0 \neq 0 \quad , \quad \text{summation in } f \text{ leads to :}$$

$$\frac{1}{1 + a \left(\frac{3N}{\pi V} \right)^{\frac{1}{3}} \left(\ln \frac{p_0^2}{\lambda} + i\pi \right)} \quad (20)$$

Using (18), the effective mass m^* is given by:

$$\frac{m}{m^2} = 1 - \frac{8}{15} (7 \ln 2 - 1) a^2 \left(\frac{3N}{\pi V} \right)^{2/3} \quad (21)$$

Similarly, the velocity of sound is given by:

$$u^2 = \frac{\pi^{4/3}}{3^{1/4}} \left(\frac{N}{V} \right)^{2/3} \frac{1}{m^2} + \frac{2\pi a}{m^2} \frac{N}{V} \left[1 + \frac{9}{15} a \left(\frac{3N}{\pi V} \right)^{\frac{1}{3}} (11 - \ln(2)) \right] \quad (22)$$

By (22), the ground state energy of the Fermi liquid can be found. From §2,

$$u^2 = \frac{N}{m} \frac{\partial \mu}{\partial N}$$

Integration (22) twice yields:

$$E = \int \mu dN = E^{(0)} + \frac{\pi a}{m} \frac{N^2}{V} \left[1 + \frac{6}{35} a \left(\frac{3N}{\pi V} \right)^{\frac{1}{3}} (11 - 2 \ln 2) \right] \quad (23)$$

Again, just as in the case for a Bose gas, we are interested in the momentum distribution of particles, for which we need to calculate the matrix element:

$$N_{p,1/2} = N_{p,-1/2} = \langle \Psi^* a_{p,1/2}^\dagger a_{p,1/2} \Psi \rangle \quad (24)$$

Using perturbation theory up to the second order:

$$\Psi_s = \Psi_s^{(0)} + \sum_{r \neq s} \frac{(H_{\text{int}})_{rs} \Psi_r^{(0)}}{E_s - E_r} + \sum_{r \neq s} \sum_{k \neq s} \frac{(H_{\text{int}})_{rk} (H_{\text{int}})_{ks} \Psi_r^{(0)}}{(E_s - E_k)(E_s - E_r)} - (H_{\text{int}})_{ss} \sum_{r \neq s} \frac{(H_{\text{int}})_{rs} \Psi_r^{(0)}}{(E_s - E_r)^2} - \frac{\Psi_s^{(0)}}{2} \sum_{r \neq s} \frac{|(H_{\text{int}})_{rs}|^2}{(E_s - E_r)^2} \quad (25)$$

By substitution of (25) into (24), we obtain:

$$N_{p,1/2} - n_{p,1/2} = \sum_{k \neq 0} \frac{|(H_{\text{int}})_{k0}|^2 (n_{p,1/2}^{(k)} - n_{p,1/2})}{(E_0 - E_k)^2} \quad (26)$$

Substitution of H_{int} from (1) into (26) yields:

$$N_{p,1/2} - n_{p,1/2} = \begin{cases} -\frac{16a^4\pi^2}{(2\pi)^6 m^2} \int_{|p_1| < p_0} dp_1 \int_{|p_2| < p_0} dp_2 \int dp_3 \frac{\delta(p+p_1-p_2-p_3)}{[(p^2+p_1^2-p_2^2-p_3^2)/2m]^2} & ; |p| < p_0 \\ \frac{16a^4\pi^2}{(2\pi)^6 m^2} \int_{|p_1| < p_0} dp_1 \int_{|p_2| < p_0} dp_2 \int dp_3 \frac{\delta(p+p_1-p_2-p_3)}{[(p^2+p_1^2-p_2^2-p_3^2)/2m]^2} & ; |p| > p_0 \end{cases} \quad (27)$$

Evaluation of (27) yields some limiting values (as noted in AGD):

$$\begin{aligned} N_{0,1/2} &= 1 - 2a^2 \left(\frac{3N}{\pi V} \right)^{2/3} \left(1 - \frac{1}{2} \ln(2) \right) \\ N_{p_0-0,1/2} &= 1 - 2a^2 \left(\frac{3N}{\pi V} \right)^{2/3} \left(\frac{1}{3} + \ln(2) \right) \\ N_{p_0+0,1/2} &= 2a^2 \left(\frac{3N}{\pi V} \right)^{2/3} \left(\ln(2) - \frac{1}{3} \right) \\ N_{p_0-0,1/2} - N_{p_0+0,1/2} &= 1 - 4a^2 \left(\frac{3N}{\pi V} \right)^{2/3} \ln(2) \\ N_{p \gg p_0,1/2} &= \frac{16a^4}{9} \left(\frac{3N}{\pi V} \right)^{2/3} \left(\frac{p_0}{p} \right)^9 \end{aligned} \quad (28)$$

For $p < p_0$, $N_p \approx 1$, & falls slowly as p increases from 0 to p_0 . N_p drops discontinuously to a value of order $a^2(N/V)^{2/3}$, & for $p > p_0$, N_p falls off according to the law:

$$a^4 \left(\frac{N}{V} \right)^{4/3} \left(\frac{p_0}{p} \right)^9 \quad (29)$$

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